

Администрирование локальных сетей

Лабораторная работа № 14. Статическая маршрутизация в Интернете.
Настройка

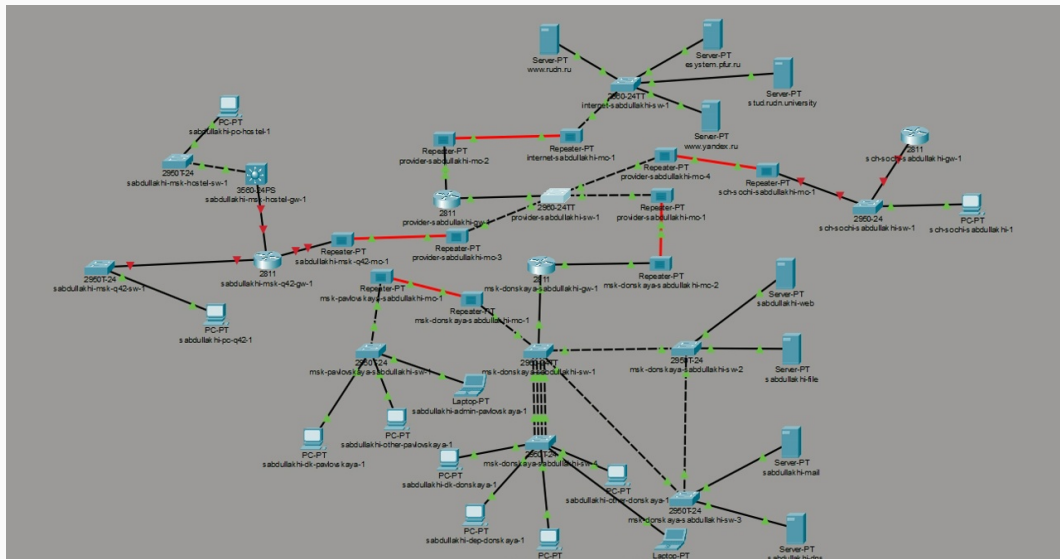
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Российский университет дружбы народов, Москва, Россия

Настроить взаимодействие через сеть провайдера посредством статической маршрутизации локальной сети организации с сетью основного здания, расположенного в 42-м квартале в Москве, и сетью филиала, расположенного в г. Сочи.

Логическая схема проекта



Настройка коммутатора provider-sw-1

```
provider-sabdullakhi-sw-1(config)#interface f0/3
provider-sabdullakhi-sw-1(config-if)#switchport mode trunk

provider-sabdullakhi-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to
down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to
up

provider-sabdullakhi-sw-1(config-if)#exit
provider-sabdullakhi-sw-1(config)#interface f0/4
provider-sabdullakhi-sw-1(config-if)#switchport mode trunk

provider-sabdullakhi-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to
down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to
up

provider-sabdullakhi-sw-1(config-if)#exit
provider-sabdullakhi-sw-1(config)#vlan 5
provider-sabdullakhi-sw-1(config-vlan)##name q42
^
% Invalid input detected at '^' marker.

provider-sabdullakhi-sw-1(config-vlan)#name q42
provider-sabdullakhi-sw-1(config-vlan)#exit
provider-sabdullakhi-sw-1(config)#interface vlan5
provider-sabdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan5, changed state to up

provider-sabdullakhi-sw-1(config-if)#no shutdown
provider-sabdullakhi-sw-1(config-if)#exit
provider-sabdullakhi-sw-1(config)#vlan 6
provider-sabdullakhi-sw-1(config-vlan)#name sochi
provider-sabdullakhi-sw-1(config-vlan)#exit
provider-sabdullakhi-sw-1(config)#interface vlan6
provider-sabdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up

provider-sabdullakhi-sw-1(config-if)#no shutdown
provider-sabdullakhi-sw-1(config-if)#exit
provider-sabdullakhi-sw-1(config)#exit
provider-sabdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

Настройка маршрутизатора msk-donskaya-gw-1

Password:

msk-donskaya-sabdullakhi-gw-1>ena

Password:

msk-donskaya-sabdullakhi-gw-1#conf term

Enter configuration commands, one per line. End with CNTL/Z.

msk-donskaya-sabdullakhi-gw-1(config)#interface f0/1.5

msk-donskaya-sabdullakhi-gw-1(config-subif)#

%LINK-3-UPDOWN: Interface FastEthernet0/1.5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up

msk-donskaya-sabdullakhi-gw-1(config-subif)#encapsulation dot1Q 5

msk-donskaya-sabdullakhi-gw-1(config-subif)#ip address 10.128.255.1
255.255.255.252

msk-donskaya-sabdullakhi-gw-1(config-subif)#description q42

msk-donskaya-sabdullakhi-gw-1(config-subif)#exit

msk-donskaya-sabdullakhi-gw-1(config)#

msk-donskaya-sabdullakhi-gw-1(config)#interface f0/1.6

msk-donskaya-sabdullakhi-gw-1(config-subif)#

%LINK-3-UPDOWN: Interface FastEthernet0/1.6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.6, changed state to up

msk-donskaya-sabdullakhi-gw-1(config-subif)#encapsulation dot1Q 6

msk-donskaya-sabdullakhi-gw-1(config-subif)#ip address 10.128.255.5
255.255.255.252

msk-donskaya-sabdullakhi-gw-1(config-subif)#description sochi

msk-donskaya-sabdullakhi-gw-1(config-subif)#

msk-donskaya-sabdullakhi-gw-1(config-subif)#exit

msk-donskaya-sabdullakhi-gw-1(config)#exit

msk-donskaya-sabdullakhi-gw-1#

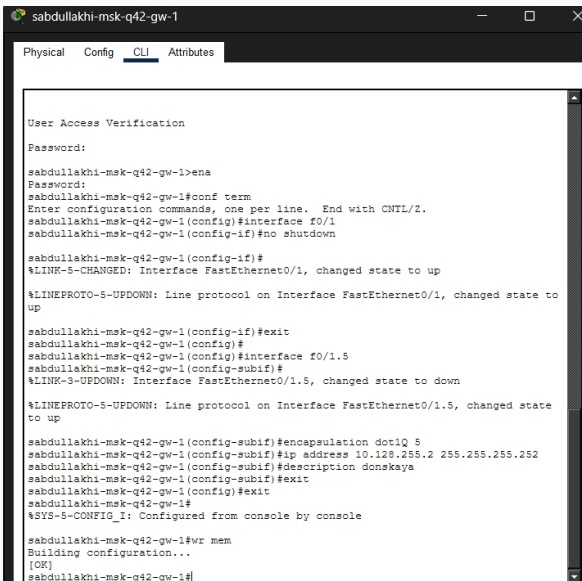
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-sabdullakhi-gw-1#wr mem

Building configuration...

[OK]

Настройка маршрутизатора msk-q42-gw-1



```
sabdullakhi-msk-q42-gw-1
Physical Config CLI Attributes

User Access Verification
Password:

sabdullakhi-msk-q42-gw-1>ena
Password:
sabdullakhi-msk-q42-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
sabdullakhi-msk-q42-gw-1(config)#interface f0/1
sabdullakhi-msk-q42-gw-1(config-if)#no shutdown

sabdullakhi-msk-q42-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to
up

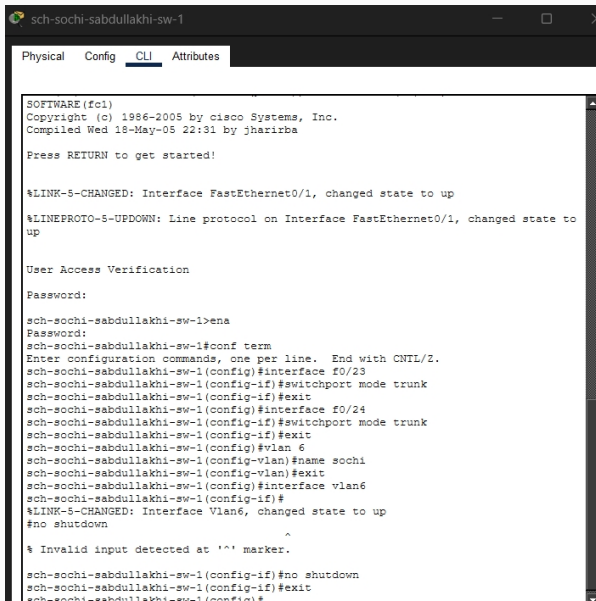
sabdullakhi-msk-q42-gw-1(config-if)#exit
sabdullakhi-msk-q42-gw-1(config)#
sabdullakhi-msk-q42-gw-1(config)#interface f0/1.5
sabdullakhi-msk-q42-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state
to up

sabdullakhi-msk-q42-gw-1(config-subif)#encapsulation dot1Q 5
sabdullakhi-msk-q42-gw-1(config-subif)#ip address 10.128.255.2 255.255.255.252
sabdullakhi-msk-q42-gw-1(config-subif)#description donskaya
sabdullakhi-msk-q42-gw-1(config-subif)#exit
sabdullakhi-msk-q42-gw-1(config)#exit
sabdullakhi-msk-q42-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sabdullakhi-msk-q42-gw-1#wr mem
Building configuration...
[OK]
sabdullakhi-msk-q42-gw-1#
```

Настройка коммутатора sch-sochi-sw-1



```
sch-sochi-sabdullakhi-sw-1
Physical Config CLI Attributes

SOFTWARE(fc1)
Copyright (c) 1986-2005 by cisco Systems, Inc.
Compiled Wed 18-May-05 22:31 by jharirba

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to
up

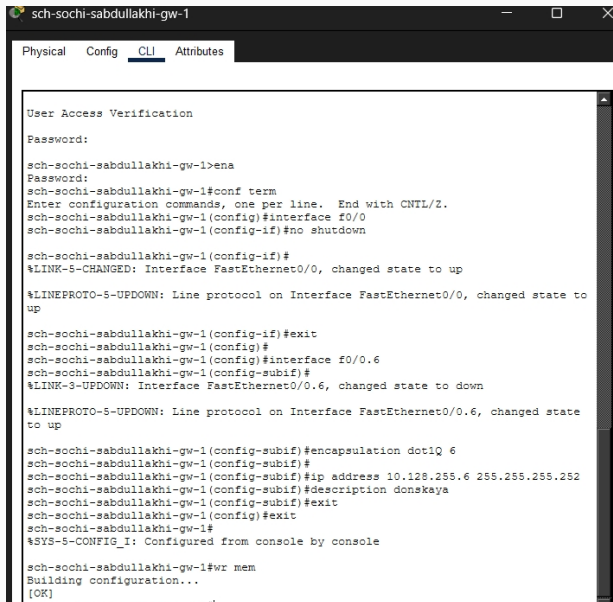
User Access Verification

Password:

sch-sochi-sabdullakhi-sw-1>ena
Password:
sch-sochi-sabdullakhi-sw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-sabdullakhi-sw-1(config)#interface f0/23
sch-sochi-sabdullakhi-sw-1(config-if)#switchport mode trunk
sch-sochi-sabdullakhi-sw-1(config-if)#exit
sch-sochi-sabdullakhi-sw-1(config)#interface f0/24
sch-sochi-sabdullakhi-sw-1(config-if)#switchport mode trunk
sch-sochi-sabdullakhi-sw-1(config-if)#exit
sch-sochi-sabdullakhi-sw-1(config)#vlan 6
sch-sochi-sabdullakhi-sw-1(config-vlan)#name sochi
sch-sochi-sabdullakhi-sw-1(config-vlan)#exit
sch-sochi-sabdullakhi-sw-1(config)#interface vlan6
sch-sochi-sabdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up
#no shutdown

sch-sochi-sabdullakhi-sw-1(config-if)#no shutdown
sch-sochi-sabdullakhi-sw-1(config-if)#exit
sch-sochi-sabdullakhi-sw-1(config)#
```

Настройка маршрутизатора sch-sochi-gw-1



The screenshot shows a terminal window titled "sch-sochi-sabdullakhi-gw-1" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying a series of commands and system messages. The configuration process starts with enabling the device, entering configuration mode, and configuring the FastEthernet0/0 interface. It then moves to the sub-interface FastEthernet0/0.6, configuring its encapsulation, IP address, and description. Finally, it saves the configuration to memory.

```
sch-sochi-sabdullakhi-gw-1>ena
Password:
sch-sochi-sabdullakhi-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-sabdullakhi-gw-1(config)#interface f0/0
sch-sochi-sabdullakhi-gw-1(config-if)#no shutdown

sch-sochi-sabdullakhi-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

sch-sochi-sabdullakhi-gw-1(config-if)#exit
sch-sochi-sabdullakhi-gw-1(config)#
sch-sochi-sabdullakhi-gw-1(config)#interface f0/0.6
sch-sochi-sabdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.6, changed state to up

sch-sochi-sabdullakhi-gw-1(config-subif)#encapsulation dot1Q 6
sch-sochi-sabdullakhi-gw-1(config-subif)#
sch-sochi-sabdullakhi-gw-1(config-subif)#ip address 10.128.255.6 255.255.255.252
sch-sochi-sabdullakhi-gw-1(config-subif)#description donskaya
sch-sochi-sabdullakhi-gw-1(config-subif)#exit
sch-sochi-sabdullakhi-gw-1(config)#exit
sch-sochi-sabdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-sabdullakhi-gw-1#wr mem
Building configuration...
[OK]
```


Настройка маршрутизатора msk-q42-gw-1

```
sabdullakhi-msk-q42-gw-1(config)#interface f0/0
sabdullakhi-msk-q42-gw-1(config-if)#no shutdown

sabdullakhi-msk-q42-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to
up

sabdullakhi-msk-q42-gw-1(config-if)#exit
sabdullakhi-msk-q42-gw-1(config)#interface f0/0.201
sabdullakhi-msk-q42-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.201, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.201, changed
state to up

sabdullakhi-msk-q42-gw-1(config-subif)#encapsulation dot1Q 201
sabdullakhi-msk-q42-gw-1(config-subif)#
sabdullakhi-msk-q42-gw-1(config-subif)#ip address 10.129.0.1 255.255.255.0
sabdullakhi-msk-q42-gw-1(config-subif)#description q42main
sabdullakhi-msk-q42-gw-1(config-subif)#exit
sabdullakhi-msk-q42-gw-1(config)#interface f1/0
sabdullakhi-msk-q42-gw-1(config-if)#no shutdown

% Invalid input detected at '^' marker.

sabdullakhi-msk-q42-gw-1(config-if)#no shutdown

sabdullakhi-msk-q42-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

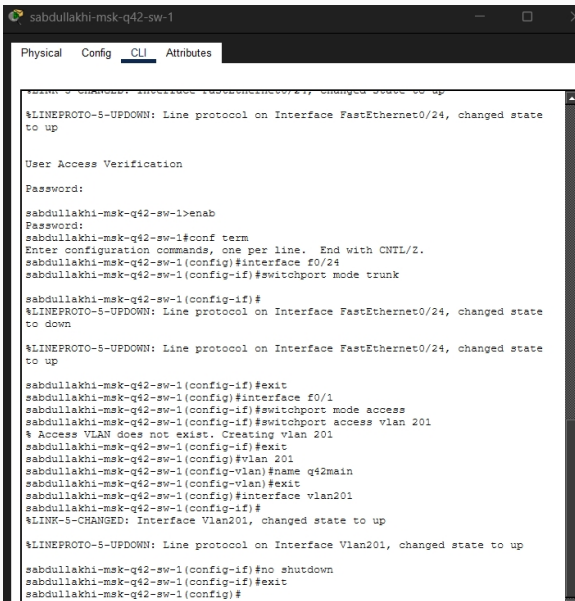
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to
up

sabdullakhi-msk-q42-gw-1(config-if)#exit
sabdullakhi-msk-q42-gw-1(config)#interface f1/0.202
sabdullakhi-msk-q42-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet1/0.202, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0.202, changed
state to up

sabdullakhi-msk-q42-gw-1(config-subif)#encapsulation dot1Q 202
sabdullakhi-msk-q42-gw-1(config-subif)#ip address 10.129.1.1 255.255.255.0
sabdullakhi-msk-q42-gw-1(config-subif)#description q42management
sabdullakhi-msk-q42-gw-1(config-subif)#exit
```

Настройка коммутатора msk-q42-sw-1



The screenshot shows a terminal window with a title bar containing the device name 'sabdullakhi-msk-q42-sw-1'. Below the title bar are four tabs: 'Physical', 'Config', 'CLI' (which is selected and highlighted), and 'Attributes'. The main area of the terminal displays the following text:

```
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

User Access Verification

Password:

sabdullakhi-msk-q42-sw-1>enab
Password:
sabdullakhi-msk-q42-sw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
sabdullakhi-msk-q42-sw-1(config)#interface f0/24
sabdullakhi-msk-q42-sw-1(config-if)#switchport mode trunk

sabdullakhi-msk-q42-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

sabdullakhi-msk-q42-sw-1(config-if)#exit
sabdullakhi-msk-q42-sw-1(config)#interface f0/1
sabdullakhi-msk-q42-sw-1(config-if)#switchport mode access
sabdullakhi-msk-q42-sw-1(config-if)#switchport access vlan 201
% Access VLAN does not exist. Creating vlan 201
sabdullakhi-msk-q42-sw-1(config-if)#exit
sabdullakhi-msk-q42-sw-1(config)#vlan 201
sabdullakhi-msk-q42-sw-1(config-vlan)#name q42main
sabdullakhi-msk-q42-sw-1(config-vlan)#exit
sabdullakhi-msk-q42-sw-1(config)#interface vlan201
sabdullakhi-msk-q42-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan201, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan201, changed state to up

sabdullakhi-msk-q42-sw-1(config-if)#no shutdown
sabdullakhi-msk-q42-sw-1(config-if)#exit
sabdullakhi-msk-q42-sw-1(config)#
```

Настройка маршрутизирующего коммутатора msk-hostel-gw-1

```
msk-hostel-gw-1
Physical Config CLI Attributes
msk-hostel-gw-1 (config)#interface g0/1
msk-hostel-gw-1 (config-if)#switchport trunk encapsulation dot1q
msk-hostel-gw-1 (config-if)#switchport mode trunk

msk-hostel-gw-1 (config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

msk-hostel-gw-1 (config-if)#exit
msk-hostel-gw-1 (config)#interface f0/1
msk-hostel-gw-1 (config-if)#switchport trunk encapsulation dot1q
msk-hostel-gw-1 (config-if)#switchport mode trunk

msk-hostel-gw-1 (config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

msk-hostel-gw-1 (config-if)#exit
msk-hostel-gw-1 (config)#vlan 202
msk-hostel-gw-1 (config-vlan)#name q42management
msk-hostel-gw-1 (config-vlan)#exit
msk-hostel-gw-1 (config)#interface vlan202
msk-hostel-gw-1 (config-if)#
%LINK-5-CHANGED: Interface Vlan202, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan202, changed state to up

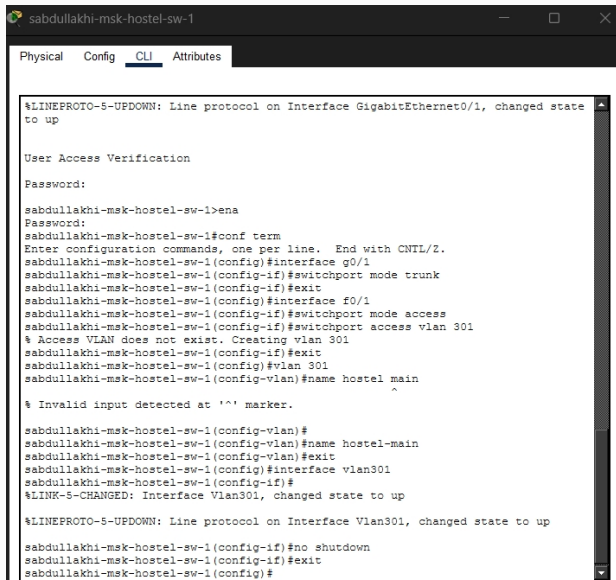
msk-hostel-gw-1 (config-if)#interface vlan202
msk-hostel-gw-1 (config-if)#no shutdown
msk-hostel-gw-1 (config-if)#ip address 10.129.1.2 255.255.255.0
msk-hostel-gw-1 (config-if)#exit
msk-hostel-gw-1 (config)#vlan 301
msk-hostel-gw-1 (config-vlan)#name hostel main
msk-hostel-gw-1 (config-vlan)#exit
% Invalid input detected at '^' marker.

msk-hostel-gw-1 (config-vlan)#name hostel-main
msk-hostel-gw-1 (config-vlan)#exit
msk-hostel-gw-1 (config)#interface vlan301
msk-hostel-gw-1 (config-if)#
%LINK-5-CHANGED: Interface Vlan301, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up

msk-hostel-gw-1 (config-if)#no shutdown
msk-hostel-gw-1 (config-if)#ip address 10.129.128.1 255.255.255.0
msk-hostel-gw-1 (config-if)#exit
```

Настройка коммутатора msk-hostel-sw-1



```
sabdullakhi-msk-hostel-sw-1
Physical Config CLI Attributes

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

User Access Verification

Password:

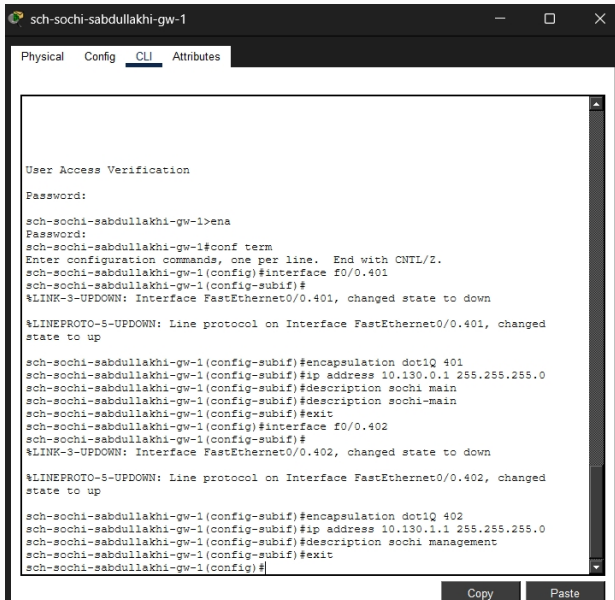
sabdullakhi-msk-hostel-sw-1>ena
Password:
sabdullakhi-msk-hostel-sw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
sabdullakhi-msk-hostel-sw-1(config)#interface g0/1
sabdullakhi-msk-hostel-sw-1(config-if)#switchport mode trunk
sabdullakhi-msk-hostel-sw-1(config-if)#exit
sabdullakhi-msk-hostel-sw-1(config)#interface f0/1
sabdullakhi-msk-hostel-sw-1(config-if)#switchport mode access
sabdullakhi-msk-hostel-sw-1(config-if)#switchport access vlan 301
% Access VLAN does not exist. Creating vlan 301
sabdullakhi-msk-hostel-sw-1(config-if)#exit
sabdullakhi-msk-hostel-sw-1(config)#vlan 301
sabdullakhi-msk-hostel-sw-1(config-vlan)#name hostel main
^
% Invalid input detected at '^' marker.

sabdullakhi-msk-hostel-sw-1(config-vlan)#
sabdullakhi-msk-hostel-sw-1(config-vlan)#name hostel-main
sabdullakhi-msk-hostel-sw-1(config-vlan)#exit
sabdullakhi-msk-hostel-sw-1(config)#interface vlan301
sabdullakhi-msk-hostel-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan301, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up

sabdullakhi-msk-hostel-sw-1(config-if)#no shutdown
sabdullakhi-msk-hostel-sw-1(config-if)#exit
sabdullakhi-msk-hostel-sw-1(config)#
```

Настройка маршрутизатора sch-sochi-gw-1



The screenshot shows a terminal window titled "sch-sochi-sabdullakhi-gw-1" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying a series of configuration commands and their outputs. The commands include enabling the router, entering configuration mode, configuring two interfaces (f0/0.401 and f0/0.402) with IP addresses, descriptions, and encapsulation. The terminal output shows the state of the interfaces changing from down to up.

```
sch-sochi-sabdullakhi-gw-1>ena
Password:
sch-sochi-sabdullakhi-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-sabdullakhi-gw-1(config)#interface f0/0.401
sch-sochi-sabdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.401, changed state to down

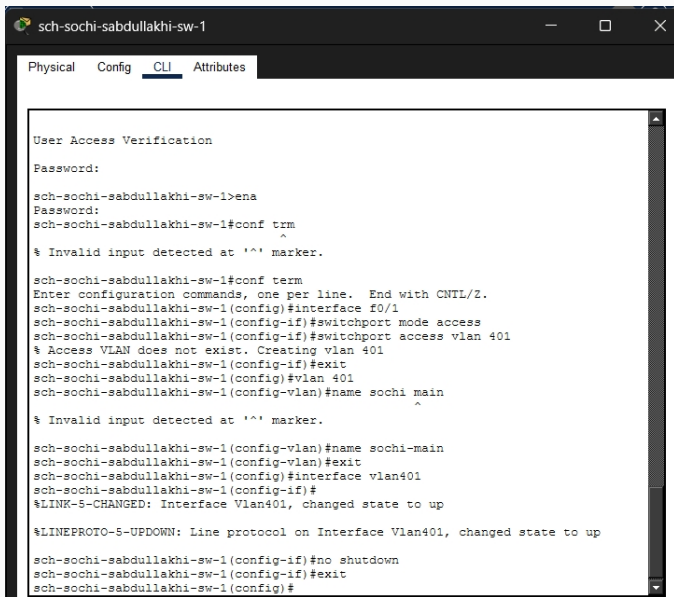
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.401, changed
state to up

sch-sochi-sabdullakhi-gw-1(config-subif)#encapsulation dot1Q 401
sch-sochi-sabdullakhi-gw-1(config-subif)#ip address 10.130.0.1 255.255.255.0
sch-sochi-sabdullakhi-gw-1(config-subif)#description sochi main
sch-sochi-sabdullakhi-gw-1(config-subif)#description sochi-main
sch-sochi-sabdullakhi-gw-1(config-subif)#exit
sch-sochi-sabdullakhi-gw-1(config)#interface f0/0.402
sch-sochi-sabdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.402, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.402, changed
state to up

sch-sochi-sabdullakhi-gw-1(config-subif)#encapsulation dot1Q 402
sch-sochi-sabdullakhi-gw-1(config-subif)#ip address 10.130.1.1 255.255.255.0
sch-sochi-sabdullakhi-gw-1(config-subif)#description sochi management
sch-sochi-sabdullakhi-gw-1(config-subif)#exit
sch-sochi-sabdullakhi-gw-1(config)#
```

Настройка коммутатора sch-sochi-sw-1



```
sch-sochi-sabdullakhi-sw-1
Physical Config CLI Attributes

User Access Verification

Password:

sch-sochi-sabdullakhi-sw-1>ena
Password:
sch-sochi-sabdullakhi-sw-1#conf trm
      ^
% Invalid input detected at '^' marker.

sch-sochi-sabdullakhi-sw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-sabdullakhi-sw-1(config)#interface f0/1
sch-sochi-sabdullakhi-sw-1(config-if)#switchport mode access
sch-sochi-sabdullakhi-sw-1(config-if)#switchport access vlan 401
% Access VLAN does not exist. Creating vlan 401
sch-sochi-sabdullakhi-sw-1(config-if)#exit
sch-sochi-sabdullakhi-sw-1(config)#vlan 401
sch-sochi-sabdullakhi-sw-1(config-vlan)#name sochi main
      ^
% Invalid input detected at '^' marker.

sch-sochi-sabdullakhi-sw-1(config-vlan)#name sochi-main
sch-sochi-sabdullakhi-sw-1(config-vlan)#exit
sch-sochi-sabdullakhi-sw-1(config)#interface vlan401
sch-sochi-sabdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan401, changed state to up

sch-sochi-sabdullakhi-sw-1(config-if)#no shutdown
sch-sochi-sabdullakhi-sw-1(config-if)#exit
sch-sochi-sabdullakhi-sw-1(config)#
```

Настройка маршрутизации между площадками

```
User Access Verification
Password:
msk-donskaya-sabdullakhi-gw-1>ena
Password:
msk-donskaya-sabdullakhi-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-sabdullakhi-gw-1(config)#ip route 10.129.0.0 255.255.0.0
10.128.255.2
msk-donskaya-sabdullakhi-gw-1(config)#ip route 10.130.0.0 255.255.0.0
10.128.255.6
msk-donskaya-sabdullakhi-gw-1(config)#
```

Рис. 13: Настройка маршрутизатора msk-donskaya-gw-1

```
sabdullakhi-msk-q42-gw-1>enab
Password:
sabdullakhi-msk-q42-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
sabdullakhi-msk-q42-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.1
sabdullakhi-msk-q42-gw-1(config)#
```

Рис. 14: Настройка маршрутизатора msk-q42-gw-1

```
sch-sochi-sabdullakhi-gw-1(config)#
sch-sochi-sabdullakhi-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.5
sch-sochi-sabdullakhi-gw-1(config)#
```

Рис. 15: Настройка маршрутизатора sch-sochi-gw-1

Настройка маршрутизации на 42-м квартале

```
sabdullakhi-msk-q42-gw-1(config)#  
sabdullakhi-msk-q42-gw-1(config)#ip route 10.129.128.0 255.255.128.0 10.129.1.2  
sabdullakhi-msk-q42-gw-1(config)#
```

Рис. 16: Настройка маршрутизатора msk-q42-gw-1

```
sabdullakhi-msk-hostel-gw-1(config)#ip routing  
sabdullakhi-msk-hostel-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.129.1.1  
sabdullakhi-msk-hostel-gw-1(config)#
```

Рис. 17: Настройка маршрутизирующего коммутатора msk-hostel-gw-1

Назначение IP-адресов конечным устройствам

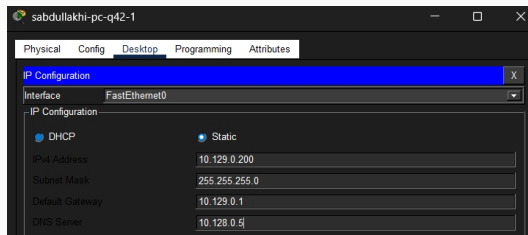
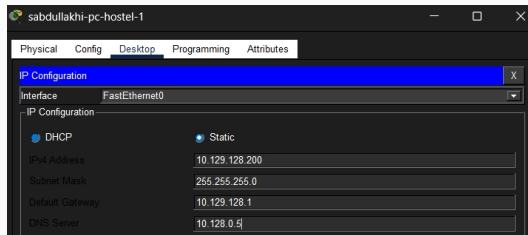


Рис. 18: Настройка ПК в 42-м квартале



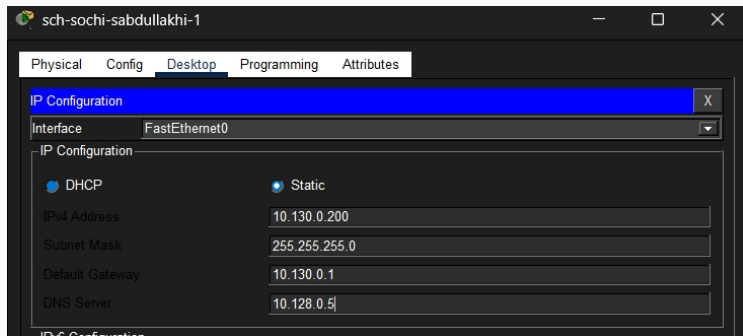


Рис. 20: Настройка ПК в Сочи

- Проверка связи между устройствами в Москве
- Трассировка маршрута до компьютера в 42-м квартале
- рассировка маршрута до компьютера в общежитии

Проверка работоспособности

```
C:\>ping 10.129.128.200

Pinging 10.129.128.200 with 32 bytes of data:

Reply from 10.129.128.200: bytes=32 time=10ms TTL=125
Reply from 10.129.128.200: bytes=32 time=10ms TTL=125
Reply from 10.129.128.200: bytes=32 time<1ms TTL=125
Reply from 10.129.128.200: bytes=32 time=10ms TTL=125

Ping statistics for 10.129.128.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 7ms

C:\>tracert 10.129.0.200

Tracing route to 10.129.0.200 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    10.128.6.1
  1  0 ms    1 ms    0 ms    10.128.255.2
  2  *        0 ms    0 ms    10.129.0.200

Trace complete.

C:\>tracert 10.129.128.200

Tracing route to 10.129.128.200 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    10.128.6.1
  1  0 ms    0 ms    0 ms    10.128.255.2
  2  0 ms    0 ms    0 ms    10.129.1.2
  3  8 ms    12 ms   0 ms    10.129.128.200
```

Просмотр таблицы маршрутизации на msk-donskaya-gw-1

```
msk-donskaya-sabdullakhi-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 198.51.100.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 18 subnets, 4 masks
C       10.128.0.0/24 is directly connected, FastEthernet0/0.3
L       10.128.0.1/32 is directly connected, FastEthernet0/0.3
C       10.128.1.0/24 is directly connected, FastEthernet0/0.2
L       10.128.1.1/32 is directly connected, FastEthernet0/0.2
C       10.128.3.0/24 is directly connected, FastEthernet0/0.101
L       10.128.3.1/32 is directly connected, FastEthernet0/0.101
C       10.128.4.0/24 is directly connected, FastEthernet0/0.102
L       10.128.4.1/32 is directly connected, FastEthernet0/0.102
C       10.128.5.0/24 is directly connected, FastEthernet0/0.103
L       10.128.5.1/32 is directly connected, FastEthernet0/0.103
C       10.128.6.0/24 is directly connected, FastEthernet0/0.104
L       10.128.6.1/32 is directly connected, FastEthernet0/0.104
C       10.128.255.0/30 is directly connected, FastEthernet0/1.5
L       10.128.255.1/32 is directly connected, FastEthernet0/1.5
C       10.128.255.4/30 is directly connected, FastEthernet0/1.6
L       10.128.255.5/32 is directly connected, FastEthernet0/1.6
S       10.129.0.0/16 [1/0] via 10.128.255.2
S       10.130.0.0/16 [1/0] via 10.128.255.6
    198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.51.100.0/28 is directly connected, FastEthernet0/1.4
L       198.51.100.2/32 is directly connected, FastEthernet0/1.4
S*    0.0.0.0/0 [1/0] via 198.51.100.1
```

Просмотр таблицы маршрутизации на msk-q42-gw-1

```
sabdullakhi-msk-q42-gw-1>enab
Password:
sabdullakhi-msk-q42-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is 10.128.255.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 7 subnets, 4 masks
C       10.128.255.0/30 is directly connected, FastEthernet0/1.5
L       10.128.255.2/32 is directly connected, FastEthernet0/1.5
C       10.129.0.0/24 is directly connected, FastEthernet0/0.201
L       10.129.0.1/32 is directly connected, FastEthernet0/0.201
C       10.129.1.0/24 is directly connected, FastEthernet1/0.202
L       10.129.1.1/32 is directly connected, FastEthernet1/0.202
S       10.129.128.0/17 [1/0] via 10.129.1.2
S*    0.0.0.0/0 [1/0] via 10.128.255.1

sabdullakhi-msk-q42-gw-1#
```

Рис. 23: Просмотр таблицы маршрутизации на msk-q42-gw-1

Просмотр таблицы маршрутизации на msk-hostel-gw-1

```
sabdullakhi-msk-hostel-gw-1>ena
Password:
sabdullakhi-msk-hostel-gw-1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is 10.129.1.1 to network 0.0.0.0

    10.0.0.0/24 is subnetted, 2 subnets
        C      10.129.1.0 is directly connected, Vlan202
        C      10.129.128.0 is directly connected, Vlan301
S*   0.0.0.0/0 [1/0] via 10.129.1.1

sabdullakhi-msk-hostel-gw-1#
```

Рис. 24: Просмотр таблицы маршрутизации на msk-hostel-gw-1

3. Вывод

В ходе выполнения лабораторной работы №14 была настроена статическая маршрутизация между тремя площадками: сетью провайдера, 42-м кварталом в Москве и филиалом в Сочи. На всех маршрутизаторах и коммутаторах выполнены необходимые настройки VLAN, транков, суб-интерфейсов и IP-адресов. Добавлены статические маршруты, включая маршруты по умолчанию, а также настроен NAT. Проверка с помощью ping и traceroute подтвердила корректную работу маршрутизации — потери пакетов отсутствуют, трассировка показывает правильный путь. Таблицы маршрутизации содержат все необходимые записи. Цель работы достигнута.

Спасибо за внимание
